

SP3172 - Report Draft 1

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Contents

Introduction	2
Methods	2
Results and Discussion	3
Conclusion	3
Future Work	3
References	3

Abstract

Extreme weather, potentially intensified by climate change, makes Singapore vulnerable to both inland (caused by excessive rainfall) and coastal floods (caused by increase in sea level). The combined effect of high speed winds over the South China Sea (SCS) and spring tides have been linked to both coastal and flash floods. These SCS winds are characteristic of the boreal winter monsoon cold surge (CS) events that occur over the Maritime Continent during the northeast (NE) monsoon season. The primary motivation of this study is to quantify the role that CS events play on extreme tide events. The study uses research-quality water level from tide gauges, obtained from the University of Hawaii Sea Level Centre from 1990 – 2013 along the Western Maritime Continent. Further analysis of the data has allowed us to quantify ‘extreme tides’ (water levels above the 99th percentile), the co-occurrences of CS events and ‘extreme tides’, along with the average increase in water levels associated with CS events. Henceforth, we develop and test a model based on water level, wind speed and pressure level anomalies, along with other climatic variables associated with CS events to predict the daily maximum and extreme water levels. The development of this model aims to improve our understanding of surge-induced flash floods, with implications for disaster preparedness in South-East Asia, thus aiding government agencies in risk mitigation for vulnerable areas and early warning systems.

Introduction

- A short overview of the monsoon season in Singapore
- What are cold surges? What is the definition of this that we are relying on?
- Potential implications and impact of the flooding season in Singapore (societal relevance) - can be a hook to the Introduction
- Gradually develop the outline with the theoretical basis of the literature (use your mid-term presentation) for reference
- Summary of the context, and then gradually scope it down to your research gap (i.e. a lack of numerical relation between cold surges and floods in Singapore): this should ideally segue into the next portion about the methods used to address the research gap in question.

Data and Methodology

- Tides have two main signals - deterministic and stochastic.
- The deterministic signals can be predicted by the phase of the celestial bodies around the Earth at that position of the tide gauge
- Removing this signal from our net tide data will allow us to obtain the pure stochastic signals - i.e water levels influenced by winds, rainfall, variation in temperature, etc.
- The stochastic signal is of concern to us.
- We tried using two forms of data - one of hourly frequency from the University of Hawaii Sea Level Centre (citation) and the other was of six minute frequency from Maritime Port Authority.
- Both data sets were processed through the UTide Package (citation).
- Reason for choosing both - MPA gives us data around singapore, while UHSLC gives us data around Peninsular Malaysia. Taking data from gives us some level of reproducibility around the area.
- Used the QA/QC manual to clean the data (cite the manual)

- After the QA/QC, the cleaned data was passed to the software through which we can separate the tidal and stochastic signals.
- Taking ERA5 re-analysis wind-speed and mean sea level pressure data, we calculate their daily area-weighted average, as has been used in Lim et al (2017) (look up citation format), and in Singapore's third climate change study V3 (citation needed) using CDO (citation needed).
- Using these processed data sets we were able to develop a multivariate linear regression model (will test other models and then check for robustness - I think this analysis should be in the next section) which allows us to predict the correlation between extreme water levels that result in nuisance flooding events and the associated weather phenomena.

Results and Discussion

Conclusion

Future Work

References